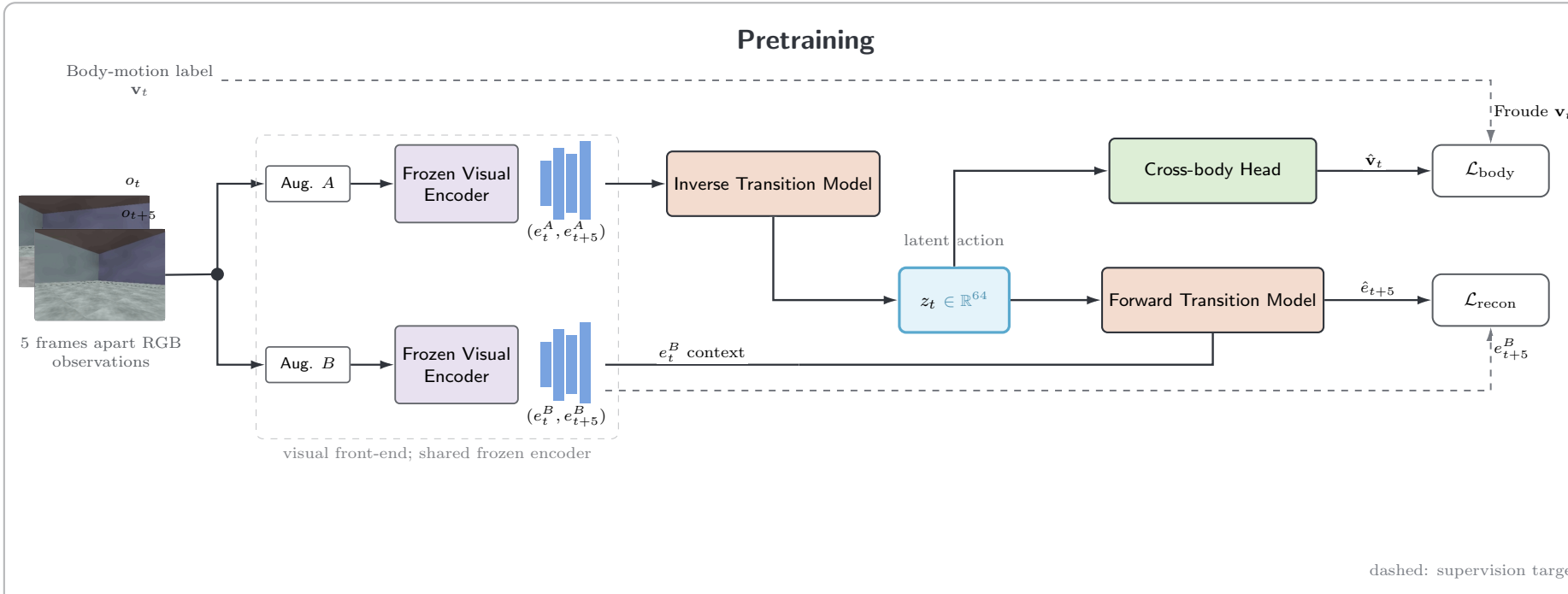

Pretraining
Body-motion label
$\mathbf{v}_t$
Froude $\mathbf{v}_t$
o_t
o_{t+5}
5 frames apart RGB observations
Aug. A
Frozen Visual Encoder
(e_t^A, e_{t+5}^A)
Aug. B
Frozen Visual Encoder
(e_t^B, e_{t+5}^B)
visual front-end; shared frozen encoder
Inverse Transition Model
latent action
$z_t \in \mathbb{R}^{64}$
Cross-body Head
$\hat{\mathbf{v}}_t$
$\mathcal{L}_{\text{body}}$
Forward Transition Model
$\hat{e}_{t+5}$
$\mathcal{L}_{\text{recon}}$
e_t^B context
e_{t+5}^B
dashed: supervision target